

3.9

b

$$f(x) = 3e^{-3x}, x > 0$$

$$E(e^{2X}) = \int_{-\infty}^{\infty} e^{2x}(f(x))dx$$

$$E(e^{2X}) = \int_0^{\infty} e^{2x}(3e^{-3x})dx$$

$$E(e^{2X}) = 3 \int_0^{\infty} e^{2x}(e^{-3x})dx$$

$$E(e^{2X}) = 3 \int_0^{\infty} e^{-x}dx$$

$$E(e^{2X}) = 3(-e^{-x}|_0^{\infty})$$

$$E(e^{2X}) = 3(0 - (-1))$$

$$E(e^{2X}) = 3$$

3.14

$$Var(X_{3.1})$$

$$Var(X_{3.1}) = E(X^2) - E(X)^2$$

$$E(X^2) = \sum_{i=1}^5 x_i^2 P(x=i)$$

$$E(X^2) = 1(1/7) + 4(1/14) + 9(3/14) + 16(2/7) + 25(2/7)$$

$$E(X^2) = \frac{197}{14}$$

$$Var(X_{3.1}) = \frac{197}{14} - E(X)^2$$

$$E(X)^2 = (\sum_{i=1}^5 x_i P(x=i))^2$$

$$E(X)^2 = (1(1/7) + 2(1/14) + 3(3/14) + 4(2/7) + 5(2/7))^2$$

$$E(X)^2 = (\frac{7}{2})^2$$

$$E(X)^2 = \frac{49}{2}$$

$$Var(X_{3.1}) = \frac{197}{14} - \frac{49}{2}$$

$$Var(X_{3.1}) = \frac{51}{28}$$

$$Var(X_{3.3})$$

$$Var(X_{3.3}) = E(X^2) - E(X)^2$$

$$E(X^2) = \int_0^\infty 3x^2 e^{-3x} dx$$

$$E(X^2) = -x^2 e^{-3x} \Big|_0^\infty + \int_0^\infty 2x e^{-3x} dx$$

$$E(X^2) = -x^2 e^{-3x} - \frac{2x e^{-3x}}{3} \Big|_0^\infty + \int_0^\infty \frac{2e^{-3x}}{3} dx$$

$$E(X^2) = -x^2 e^{-3x} - \frac{2x e^{-3x}}{3} - \frac{2e^{-3x}}{9} \Big|_0^\infty$$

$$E(X^2) = -\frac{(9x^2 + 6x + 2)e^{-3x}}{9} \Big|_0^\infty$$

$$E(X^2) = \frac{2}{9}$$

$$Var(X_{3.3}) = \frac{2}{9} - E(X)^2$$

$$E(X)^2 = (\int_0^\infty 3x e^{-3x} dx)^2$$

$$E(X)^2 = \left(-\left(xe^{-3x} + \frac{e^{-3x}}{3}\right)\Big|_0^\infty\right)^2$$

$$E(X)^2 = \left(\frac{1}{3}\right)^2$$

$$E(X)^2 = \frac{1}{9}$$

$$Var(X_{3.3}) = \frac{2}{9} - \frac{1}{9} = \frac{1}{9}$$

3.15

(a)

$$E(3X + 2) = 11$$

(b)

$$E(X^2) = E(X)^2 + Var(X) = 7$$

3.18

a

$$P(2 < X < 6) = -\Phi(-0.5) + \Phi(1.5)$$

$$P(2 < X < 6) = -0.3085 + 0.9332$$

$$P(2 < X < 6) = 0.6247$$

b

$$1 - \Phi(z) = 0.33$$

$$\Phi(z) = 0.66$$

$$z = 0.42$$

$$c = 3 + 2z$$

$$c = 3 + 0.84$$

$$c = 3.84$$

c

$$E(X^2) = E(X)^2 - Var(X) = 9 - 4 = 5$$

3.32

(d)

$$E(X^{\frac{1}{4}}) = \frac{1}{2} \int_1^{\infty} x^{\frac{1}{4}} x^{-\frac{3}{2}} dx$$

$$E(X^{\frac{1}{4}}) = \frac{1}{2} \int_1^{\infty} x^{-\frac{5}{4}} dx$$

$$E(X^{\frac{1}{4}}) = \frac{1}{2} \frac{-4}{x^{\frac{1}{4}}} \Big|_1^{\infty}$$

$$E(X^{\frac{1}{4}}) = 2$$

3.68

a

$$E(Z^4) = \frac{1}{\sqrt{2\pi}} \int x^4 e^{-\frac{x^2}{2}} dx$$

$$E(Z^4) = \frac{1}{\sqrt{2\pi}} \int x^3 (x e^{-\frac{x^2}{2}}) dx$$

$$E(Z^4) = \frac{1}{\sqrt{2\pi}} (-x^3 e^{-x^2/2} |_{-\infty}^{\infty} - \int -3x^2 e^{-\frac{x^2}{2}} dx)$$

Due to the odd function rule the first half of the integral goes to 0.

$$E(Z^4) = \frac{1}{\sqrt{2\pi}} (0 - \int -3x^2 e^{-\frac{x^2}{2}} dx)$$

$$E(Z^4) = 3 \left(\int \frac{1}{\sqrt{2\pi}} x^2 e^{-\frac{x^2}{2}} dx \right)$$

$$E(Z^4) = 3E(Z^2) = 3(1) = 3$$

b

$$Z = \frac{X-\mu}{\sigma} \rightarrow \sigma Z + \mu = X$$

$$E(X^4) = E((\sigma Z + \mu)^4)$$

$$E(X^4) = E((\sigma Z)^4 + 4\mu(\sigma Z)^3 + 6\mu^2(\sigma Z)^2 + 4\mu^3\sigma Z + \mu^4)$$

$$E(X^4) = \sigma^4(E(Z^4)) + 4\mu\sigma^3(E(Z^3)) + 6\mu^2\sigma^2(E(Z^2)) + 4\mu^3\sigma E(Z) + \mu^4$$

$$E(X^4) = \sigma^4(3) + 4\mu\sigma^3(0) + 6\mu^2\sigma^2(1) + 4\mu^3\sigma(0) + \mu^4$$

$$E(X^4) = 3\sigma^4 + 6\mu^2\sigma^2 + \mu^4$$

3.71

$$\sqrt{6}z = 5$$

$$z = \frac{5}{\sqrt{6}}$$

$$\Phi\left(\frac{5}{\sqrt{6}}\right) = \Phi(2.04) = 0.9793$$

$$P(X > 12 : 05) = 1 - \Phi(2.04) = 0.0207$$

4.20

$$H \sim \text{Bin}(10000, 0.5)$$

$$H \sim N(E(H), \text{Var}(H))$$

$$E(H) = 10000 * 0.5 = 5000$$

$$\text{Var}(H) = 10000 * 0.5 * 0.5 = 2500$$

$$H \sim N(5000, 2500)$$

$$\sigma = 500$$

$$100 = 500z \rightarrow z = 0.2$$

$$P(4900 < H < 5100) = \Phi(0.2) - \Phi(-0.2)$$

$$P(4900 < H < 5100) = 0.5793 - 0.4207 = 0.1586$$

$$\text{Alternatively, } P(4900 < H < 5100) = 2\Phi(0.2) - 1 = 0.1586$$